CSE 400: Fundamentals of Probability in Computing
Milestone 3 – Project Scribe

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Scribe Question 1: System Objective and Deterministic Baseline

In our project we plan to analyze and examine the Variance Risk Premium (VRP) and understand how it can be used to explain or predict future stock market behavior. In Milestone 1, we explored the conceptual idea of the research paper *Expected Stock Returns and Variance Risk Premia*. The paper suggests that the difference between implied volatility and realized volatility contains important information about market risk and expected returns.

In Milestone 2, this idea was extended by introducing a probabilistic framework. In that stage, true market volatility was treated as a latent random variable, meaning that it cannot be directly observed in financial markets. Instead, we use implied variance (IV) and realized variance (RV) as observable estimates of volatility. However, both of these measures may contain noise due to measurement errors or market imperfections.

Milestone 3 focuses on computational implementation and simulation. The main goal is to compute the Variance Risk Premium using real market data and evaluate how measurement errors in volatility estimation can affect the final results.

The deterministic baseline used in this project is the standard VRP formula that is widely used in financial studies. The variance risk premium is defined as the difference between squared implied volatility and squared realized volatility:

$$VRP = IV^2 - RV^2$$

Here,

- IV represents implied volatility obtained from the VIX index
- RV represents realized volatility calculated from historical S&P 500 returns

This model is considered deterministic because the calculation follows a fixed mathematical formula without any randomness in the decision process. For any given values of implied volatility and realized volatility, the VRP is computed directly using the same equation.

This deterministic approach is commonly used in financial research because it provides a simple and clear method for estimating the variance risk premium. It helps researchers compare market expectations of volatility with the actual volatility observed in the market. By starting with this deterministic baseline, we can later analyze how measurement errors influence the calculation and explore improved probabilistic models.

Scribe Question 2: Random Variables in the Simulation

In our system, several variables are uncertain and are therefore modeled as random variables. These variables represent different aspects of market volatility and financial risk. The same variables were introduced conceptually in Milestone 1 and later modeled probabilistically in Milestone 2.

Implied Volatility

Implied volatility represents the market's expectation of future volatility and is obtained from the VIX index. In our model, implied volatility is calculated as

$$IV = \frac{VIX}{100}$$

This conversion transforms the VIX percentage value into decimal form so that mathematical calculations can be performed correctly.

Even though implied volatility is derived from market data, it still contains uncertainty because option prices are influenced by market sentiment, liquidity, and other economic factors.

Realized Volatility

Realized volatility represents the actual volatility observed in the stock market. It is calculated using historical return data from the S&P 500 index. Similar to implied volatility, realized volatility values are also converted into decimal form:

$$RV = \frac{RV\%}{100}$$

Since realized volatility is estimated from historical data, it may contain estimation errors caused by sampling limitations or market microstructure effects.

Measurement Error Terms

To capture uncertainty in volatility estimation, the corrected model introduces measurement error terms:

$$\epsilon_{IV}, \quad \epsilon_{RV}$$

These variables represent the estimation errors present in implied volatility and realized volatility.

For simplicity, we assume that these errors have:

- Mean approximately equal to zero
- Relatively small variance

This means that the errors fluctuate randomly around the true values without introducing systematic bias.

Variance Risk Premium

The variance risk premium itself is also a random variable because it depends on uncertain volatility estimates. The standard VRP formula is

$$VRP = IV^2 - RV^2$$

Since both IV and RV contain measurement noise, the calculated VRP can also vary depending on the estimation errors.

Future Market Behavior

Although this milestone mainly focuses on computing VRP, the long-term objective of the project is to understand how VRP relates to future market behavior and how investors are compensated for taking volatility risk.

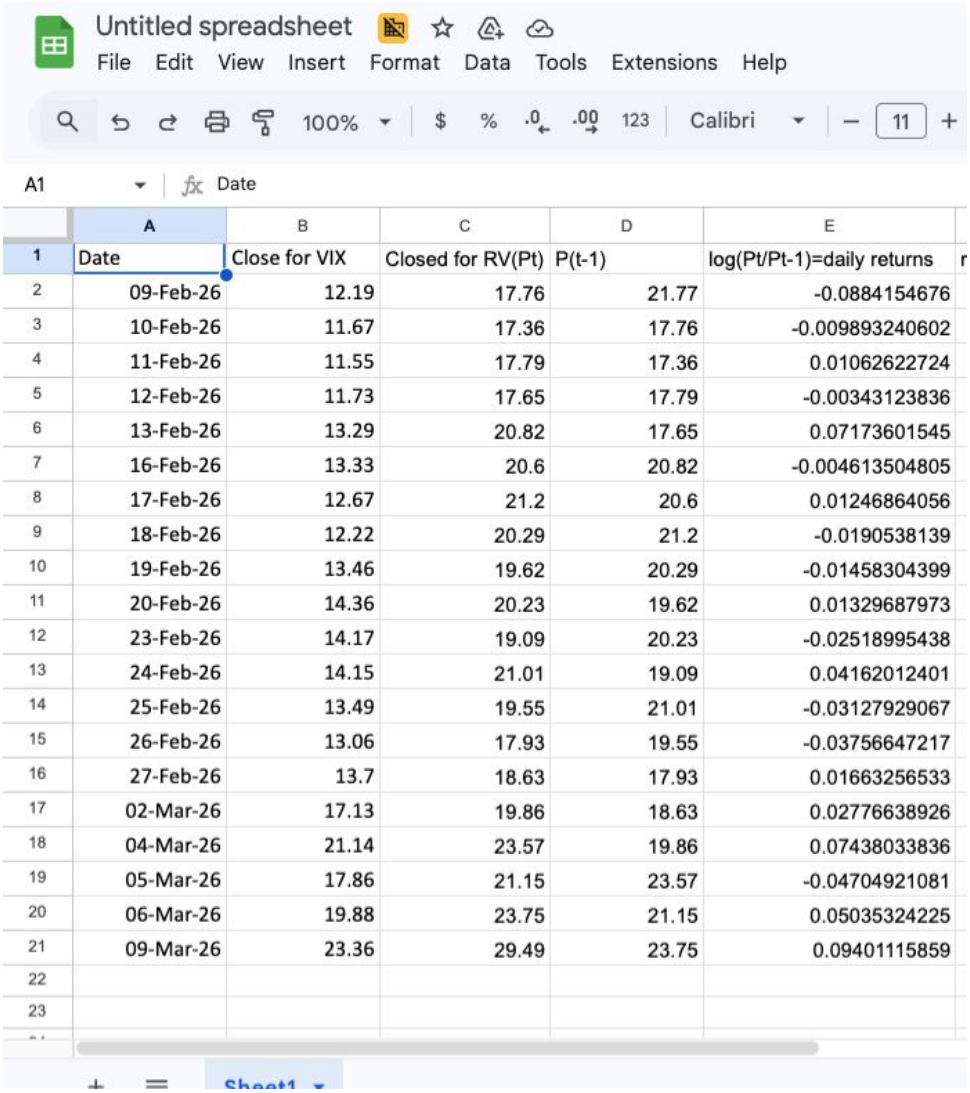
Scribe Question 3: Probabilistic Modeling and Fixed Decision Rule

In this simulation, uncertainty is modeled by treating market data as an imperfect reflection of reality. We assume that the observed values for Implied Volatility (IV) and Realized Volatility (RV) are not absolute truths but are "noisy" proxies.

- **Probabilistic Assumptions:** We assume there exists a latent "True Volatility" (V_{true}). To simulate real-world data imperfections, we introduce stochastic error terms, ϵ_{IV} and ϵ_{RV}. These represent measurement noise such as reporting lags, bid-ask spreads, or sampling errors.
- **Deterministic Baseline:** While the inputs are probabilistic (uncertain), our decision rule remains deterministic. The model consistently applies the Variance Risk Premium (VRP) formula regardless of the noise level. This allows us to test if the VRP signal is robust enough to survive observational errors.

Mathematical Framework: The error-adjusted VRP is calculated as follows:

$$\mathrm{IV}^2_{\text{final}} - \mathrm{RV}^2_{\text{final}} - (\mathrm{IV}_{\text{error}} - \mathrm{RV}_{\text{error}})$$



	A	B	C	D	E
1	Date	Close for VIX	Closed for RV(Pt)	P(t-1)	log(Pt/Pt-1)=daily returns
2	09-Feb-26	12.19	17.76	21.77	-0.0884154676
3	10-Feb-26	11.67	17.36	17.76	-0.009893240602
4	11-Feb-26	11.55	17.79	17.36	0.01062622724
5	12-Feb-26	11.73	17.65	17.79	-0.00343123836
6	13-Feb-26	13.29	20.82	17.65	0.07173601545
7	16-Feb-26	13.33	20.6	20.82	-0.004613504805
8	17-Feb-26	12.67	21.2	20.6	0.01246864056
9	18-Feb-26	12.22	20.29	21.2	-0.0190538139
10	19-Feb-26	13.46	19.62	20.29	-0.01458304399
11	20-Feb-26	14.36	20.23	19.62	0.01329687973
12	23-Feb-26	14.17	19.09	20.23	-0.02518995438
13	24-Feb-26	14.15	21.01	19.09	0.04162012401
14	25-Feb-26	13.49	19.55	21.01	-0.03127929067
15	26-Feb-26	13.06	17.93	19.55	-0.03756647217
16	27-Feb-26	13.7	18.63	17.93	0.01663256533
17	02-Mar-26	17.13	19.86	18.63	0.02776638926
18	04-Mar-26	21.14	23.57	19.86	0.07438033836
19	05-Mar-26	17.86	21.15	23.57	-0.04704921081
20	06-Mar-26	19.88	23.75	21.15	0.05035324225
21	09-Mar-26	23.36	29.49	23.75	0.09401115859
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A1 | Date

	G	H	I	J	K	L	M	N	O	P	Q
1	RV	IV=VIX/100	IV*IV	RV*RV	VRP=IV^2-RV^2	true volatality	IV_error	RV_error	Final_IV	Final_RV	Final_VRP
2	0.1776	0.2177	0.04739329	0.03154176	0.01585153	0.19765	0.02005	-0.02005	0.19765	0.19765	-0.0401
3	0.1736	0.1776	0.03154176	0.03013696	0.0014048	0.1756	0.002	-0.002	0.1756	0.1756	-0.004
4	0.1779	0.1736	0.03013696	0.03164841	-0.00151145	0.17575	-0.00215	0.00215	0.17575	0.17575	0.0043
5	0.1765	0.1779	0.03164841	0.03115225	0.00049616	0.1772	0.0007	-0.0007	0.1772	0.1772	-0.0014
6	0.2082	0.1765	0.03115225	0.04334724	-0.01219499	0.19235	-0.01585	0.01585	0.19235	0.19235	0.0317
7	0.206	0.2082	0.04334724	0.042436	0.00091124	0.2071	0.0011	-0.0011	0.2071	0.2071	-0.0022
8	0.212	0.206	0.042436	0.044944	-0.002508	0.209	-0.003	0.003	0.209	0.209	0.006
9	0.2029	0.212	0.044944	0.04116841	0.00377559	0.20745	0.00455	-0.00455	0.20745	0.20745	-0.0091
10	0.1962	0.2029	0.04116841	0.03849444	0.00267397	0.19955	0.00335	-0.00335	0.19955	0.19955	-0.0067
11	0.2023	0.1962	0.03849444	0.04092529	-0.00243085	0.19925	-0.00305	0.00305	0.19925	0.19925	0.0061
12	0.1909	0.2023	0.04092529	0.03644281	0.00448248	0.1966	0.0057	-0.0057	0.1966	0.1966	-0.0114
13	0.2101	0.1909	0.03644281	0.04414201	-0.0076992	0.2005	-0.0096	0.0096	0.2005	0.2005	0.0192
14	0.1955	0.2101	0.04414201	0.03822025	0.00592176	0.2028	0.0073	-0.0073	0.2028	0.2028	-0.0146
15	0.1793	0.1955	0.03822025	0.03214849	0.00607176	0.1874	0.0081	-0.0081	0.1874	0.1874	-0.0162
16	0.1863	0.1793	0.03214849	0.03470769	-0.0025592	0.1828	-0.0035	0.0035	0.1828	0.1828	0.007
17	0.1986	0.1863	0.03470769	0.03944196	-0.00473427	0.19245	-0.00615	0.00615	0.19245	0.19245	0.0123
18	0.2357	0.1986	0.03944196	0.05555449	-0.01611253	0.21715	-0.01855	0.01855	0.21715	0.21715	0.0371
19	0.2115	0.2357	0.05555449	0.04473225	0.01082224	0.2236	0.0121	-0.0121	0.2236	0.2236	-0.0242
20	0.2375	0.2115	0.04473225	0.05640625	-0.011674	0.2245	-0.013	0.013	0.2245	0.2245	0.026
21	0.2949	0.2375	0.05640625	0.08696601	-0.03055976	0.2662	-0.0287	0.0287	0.2662	0.2662	0.0574
22											

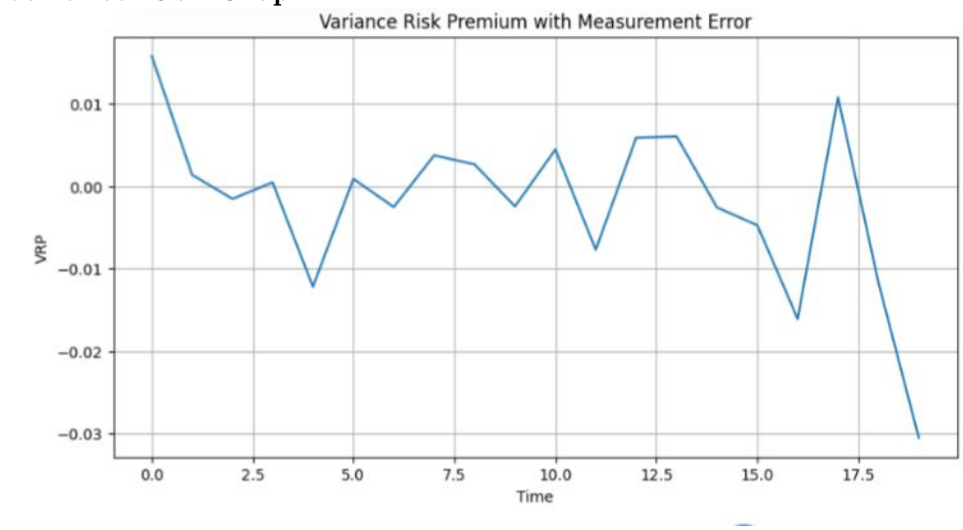
Question-4: Model Implementation Alignment

Describe how the current probabilistic model aligns with the project's implementation or experimental setup. Highlight any assumptions that influence design or evaluation choices

Solution: The current model implementation aligns with the project's goal of addressing flaws in traditional volatility research. Specifically, it addresses the unrealistic assumption that market data is error-free.

By explicitly programming columns for `IV_error` and `RV_error` in the Python simulation, the experimental setup mirrors the messy reality of financial markets. This alignment influences our evaluation choices: instead of seeking a single "correct" value, we analyze the trend and stability of the VRP signal. The model allows us to observe whether the "Risk Premium" (investors paying more for protection) remains a visible phenomenon even when the data is intentionally degraded with noise.

Variance Risk Graph:



Question-5: Cross Milestone Consistency and Change

Describe the current state of your project's probabilistic model. Identify the assumptions, components, or relationships that are presently well defined and indicate which aspects are expected to evolve or require refinement in subsequent milestones. Briefly justify your reasoning.

Solution: The project is currently in a stable baseline phase where the fundamental data processing is well-defined, but the predictive application remains an area for growth.

- **Well-Defined Components:** The conversion of raw prices into Log Returns and the core logic of the VRP relationship ($IV^2 - RV^2$) are now fully established and consistent across our simulation milestones.
- **Expected Evolution:** As seen in the current output graph, the VRP occasionally dips into negative territory. In subsequent milestones, we expect to refine how we estimate the latent "True Volatility" to distinguish between actual market signals and simulated noise. We will move from simply calculating these values to using them in a regression model to determine their predictive power for future stock returns.

Reasoning: This evolution is necessary because while we can now measure the premium under noisy conditions, the ultimate goal is to use this "cleaned" signal to forecast market movements.

Question-6: OpenIssues and Responsibility Attributions

Identify any unresolved probabilistic questions or ambiguities in the project at this stage and indicate which roles or tasks are responsible for addressing them in the next milestone.

Solution: Even though the project already computes the Variance Risk Premium, there are still some open issues that need to be addressed. One of the main uncertainties is how measurement errors in implied and realized volatility should be modeled. At the moment, the project assumes that these errors are small and randomly distributed, but this assumption has not been fully tested. Another issue is the size of the dataset used in the analysis. Since the current dataset contains only a limited number of daily observations, the results may not fully capture long-term market behavior. Increasing the dataset size could help improve the reliability of the findings. There is also the question of how best to evaluate the corrected VRP model. While basic graphs and comparisons provide useful insights, additional statistical analysis may be needed to determine whether the corrected model produces more accurate estimates. In the next milestone, the main tasks will include collecting a larger dataset, improving the modeling of measurement errors, and performing deeper statistical analysis of the VRP results. These steps will help strengthen the overall model and make the conclusions more reliable.